THE CATEGORY OF COHERENT SHEAVES WITH AN I-ADIC FILTRATION STRUCTURE ON A NOETHERIAN ALGEBRAIC SPACE IS ABELIAN, WITH EXACTNESS ETALE-LOCAL AND FLAT-PULLBACK EXACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9562-9574 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9576-9612, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The schemes chapter carries a same-named lemma used as input; it lies in Cohomology of Schemes, outside both pools, and its card status (if any) is another lane's.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Scheme-side theory of inverse systems of coherent modules (Cohomology of Schemes, lemmas inverse-systems-abelian, inverse-systems-pullback), etale descent for coherent data as in the chapter's preceding discussion, and the groupoid-descent formal template (Groupoids, lemma-abelian).

Source reference. Original lemma 087X: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9612–9612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0887: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9612–9612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 077S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9612–9612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9612–9612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals.*

(1) *The category $Coh(X, \mathcal{I})$ is abelian.*
(2) *Exactness in $Coh(X, \mathcal{I})$ can be checked étale locally.*
(3) *For any flat morphism $f : X' \to X$ of Noetherian algebraic spaces the functor $f^* : Coh(X, \mathcal{I}) \to Coh(X', f^{-1}\mathcal{I}\mathcal{O}_{X'})$ is exact.*

Proof. Proof of (1). Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X$. Set $U_1 = U_0 \times_X U_0$ and $U_2 = U_0 \times_X U_0 \times_X U_0$ as in our discussion of étale descent above. The categories $Coh(U_i, \mathcal{I}_i)$ are abelian (Cohomology of Schemes, Lemma 087X) and the pullback functors are exact functors $Coh(U_0, \mathcal{I}_0) \to Coh(U_1, \mathcal{I}_1)$ and $Coh(U_1, \mathcal{I}_1) \to Coh(U_2, \mathcal{I}_2)$ (Cohomology of Schemes, Lemma 0887). The lemma then follows formally from the description of $Coh(X, \mathcal{I})$ as a category of descent data. Some details omitted; compare with the proof of Groupoids, Lemma 077S.

Part (2) follows immediately from the discussion in the previous paragraph. In the situation of (3) choose a commutative diagram

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

where U' and U are affine schemes and the vertical morphisms are surjective étale. Then $U' \to U$ is a flat morphism of Noetherian schemes (Morphisms of Spaces, Lemma 03MM) whence the pullback functor $Coh(U, \mathcal{I}\mathcal{O}_U) \to Coh(U', \mathcal{I}\mathcal{O}_{U'})$ is exact by Cohomology of Schemes, Lemma 0887. Since we can check exactness in $Coh(X, \mathcal{O}_X)$ on U and similarly for X', U' the assertion follows. □